SignalLab: Digital Communication Toolkit

ITT 036 — Programming Assignment 1 — Project Specification Report

Enrollment Numbers	2024BITE036 | 2024BITE050 | 2024BITE058
Language	Java (JDK 17 or later)
Libraries	Java standard library, java.util, java.awt, javax.swing
Interface	Command-line menu with Java Swing waveform plots

1. Objective

SignalLab is a Java application that demonstrates how digital information can be represented and transmitted using line-coding, scrambling, and analog-to-digital modulation techniques. It accepts a digital bitstream or sampled analog input and displays the processed data and signal waveform.

2. Main Features

• Data generation and analysis: generates random binary sequences, sequences containing four or eight consecutive zeros, and supports user-entered binary data. It also finds a longest palindromic substring.
• Line coding: NRZ-L, NRZ-I, Manchester, Differential Manchester, and Alternate Mark Inversion (AMI).
• Scrambling: B8ZS and HDB3 scrambling techniques, with descrambling support.
• Modulation: Pulse Code Modulation (PCM) and Delta Modulation (DM), with decoding or reconstruction routines.
• Decoding and visualization: decodes supported line codes and displays signal waveforms using a Java graphical interface.

3. Program Flow

The user selects digital input or analog samples. For digital input, the user can generate a bitstream or enter a custom binary sequence, then choose a line-coding method. If AMI is selected, scrambling can optionally be applied. For analog input, the user selects PCM or DM and supplies sample values. The application displays the resulting data and waveform, and decoder functions can be used to recover the bitstream or reconstruct sample values.

4. Assumptions and Limitations

• Digital input contains only 0 and 1; generated sequence length is between 8 and 10,000 bits.
• Analog samples are assumed to be normalized to the range −1.0 to 1.0; PCM supports 1–16 bits per sample.
• A desktop environment is required for Swing waveform plots.
• The project is intended for educational demonstration and does not transmit signals through physical communication hardware.
Exact behavior and edge cases should be verified by running the program.

5. How to Compile and Run

Install JDK 17 or later and open a terminal in the project directory. On Linux/macOS, compile the source files with:

```
javac -d bin $(find src -name '*.java')
```

Run the application with:

```
java -cp bin Main
```

If included in the project, run the verification checks with:

```
java -cp bin ProjectTest
```

On Windows, compile the Java source files under `src` using PowerShell or a Java IDE.

6. References and Acknowledgement

1. ITT 036, “Programming Assignment 1: Implement a Digital Signal Encoder and Decoder,” Autumn 2026, assignment handout dated 21 September 2026.
2. Oracle, Java Platform, Standard Edition Documentation: https://docs.oracle.com/en/java/.
This report was prepared using the supplied assignment brief and project information. Any copied code, AI assistance, or external help should be acknowledged according to course requirements.